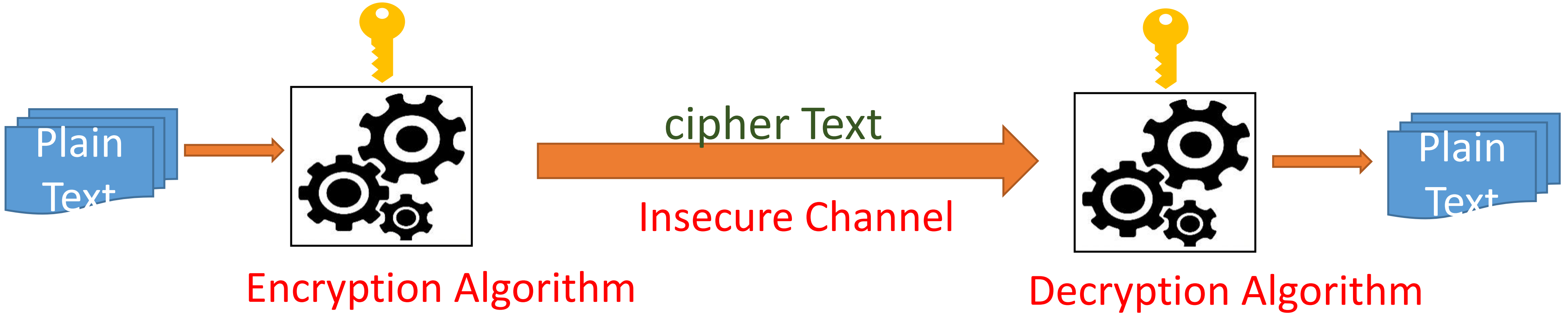
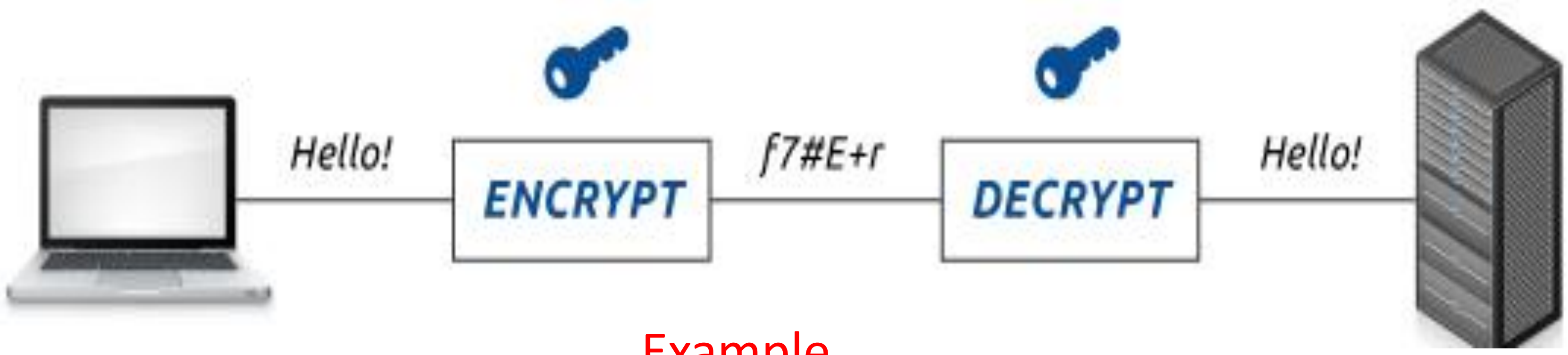
Introduction to cryptography

Module I

Why cryptography is Important

Cryptography is the science of keeping the data secure from unwanted/untrusted entities. (Highly relevant in the context of current digital world).





Example

Cryptography is the art of science encompassing the principles and methods of transforming an Intelligible message into one that is unintelligible, and then retransforming that message back to its original form.

Introduction: Security Overview

- Secure transmission of data is paramount importance.
- The major requirements of security is Confidentiality (C), Integrity (I), Availability (A). Besides these requirements, authenticity and Non-repudiation are also Important.

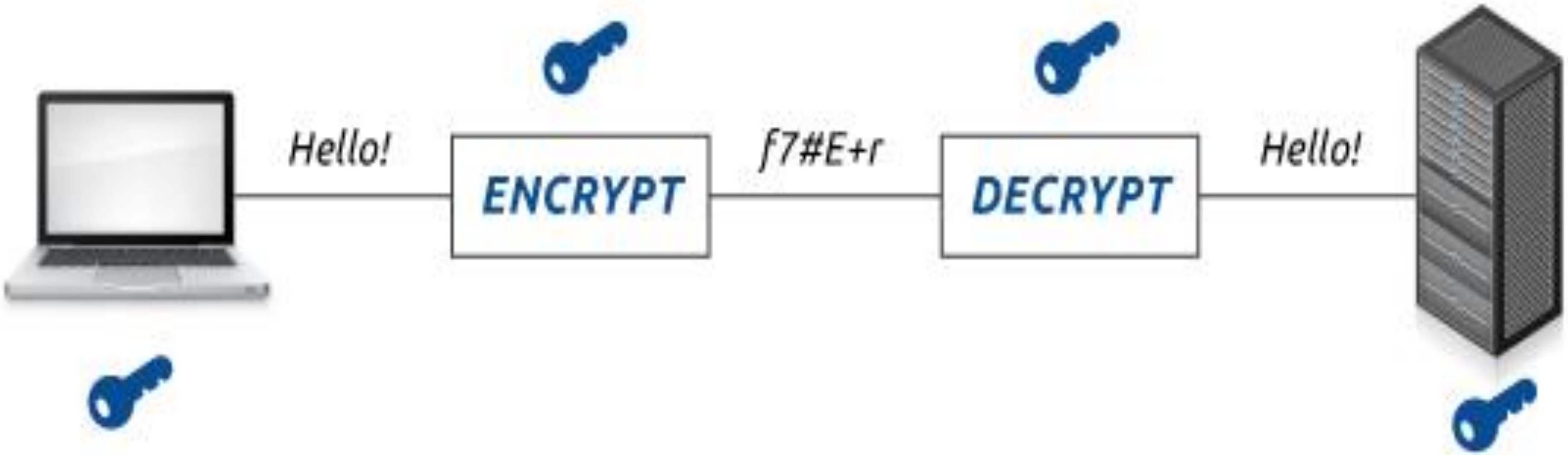
CIA triad

Security Overview - cntd..

Security requirement	Security technique
Confidentiality	Encryption
Authentication and Non-repudiation	Public Key cryptographic techniques
Integrity	Digital Signature Algorithms

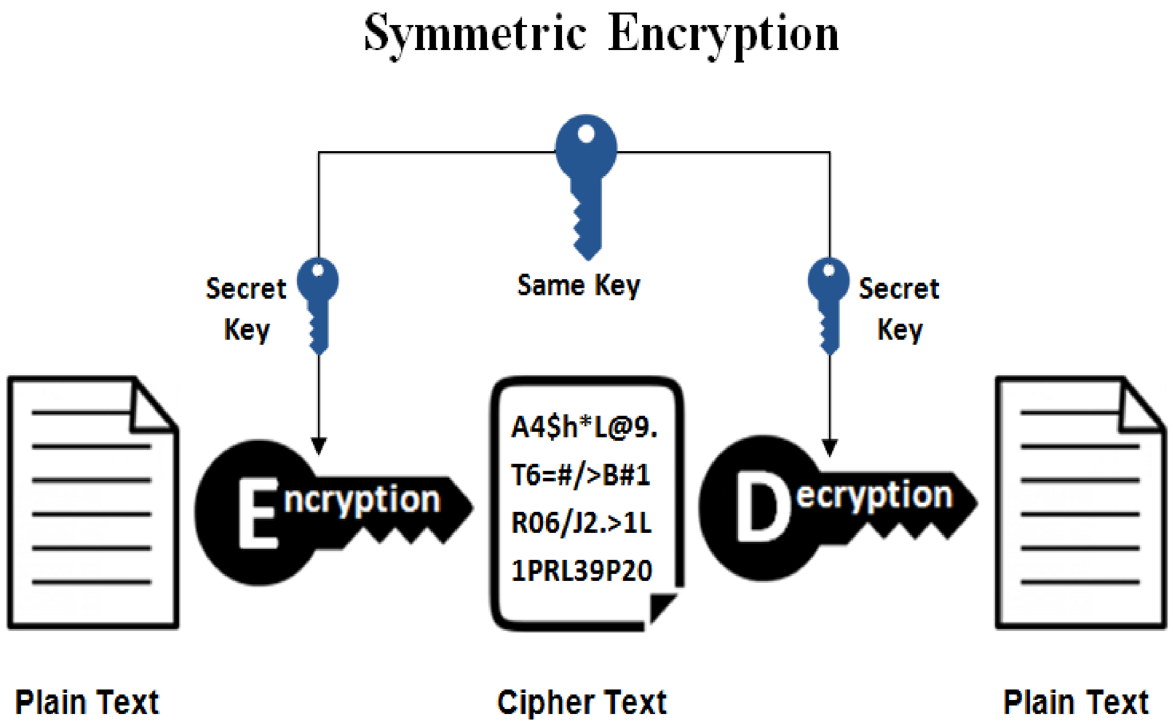
Data Confidentiality: Encryption

- Data confidentiality can be achieved by encryption.

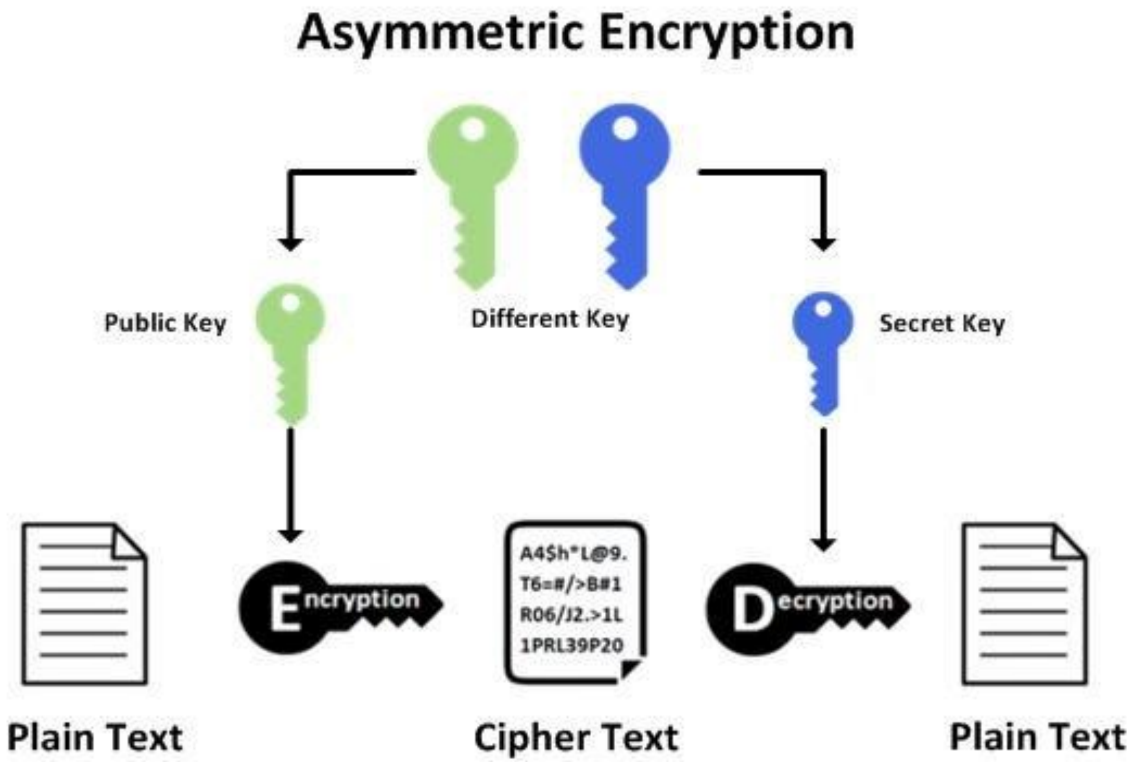


Example encryption algorithms: DES-56, AES (128, 192 and 256 bit key sizes), RSA (1024, 2048 .. Bit key sizes), ECC (Elliptic Curve Cryptography)

Encryption: Symmetric Vs Asymmetric



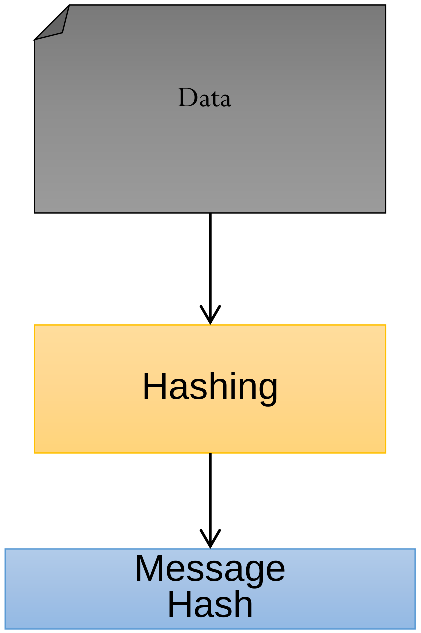
Eg. DES, 3DES, AES, RC5



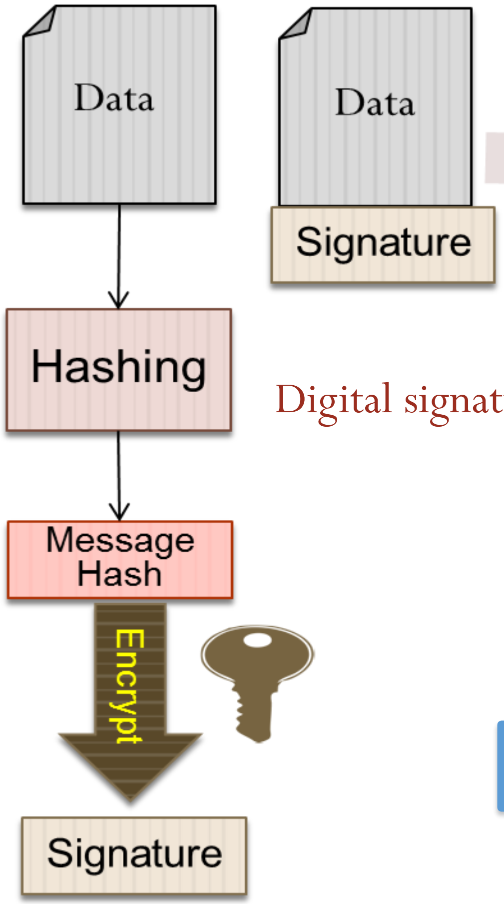
Eg. RSA, ECC

Public key and private keys are related

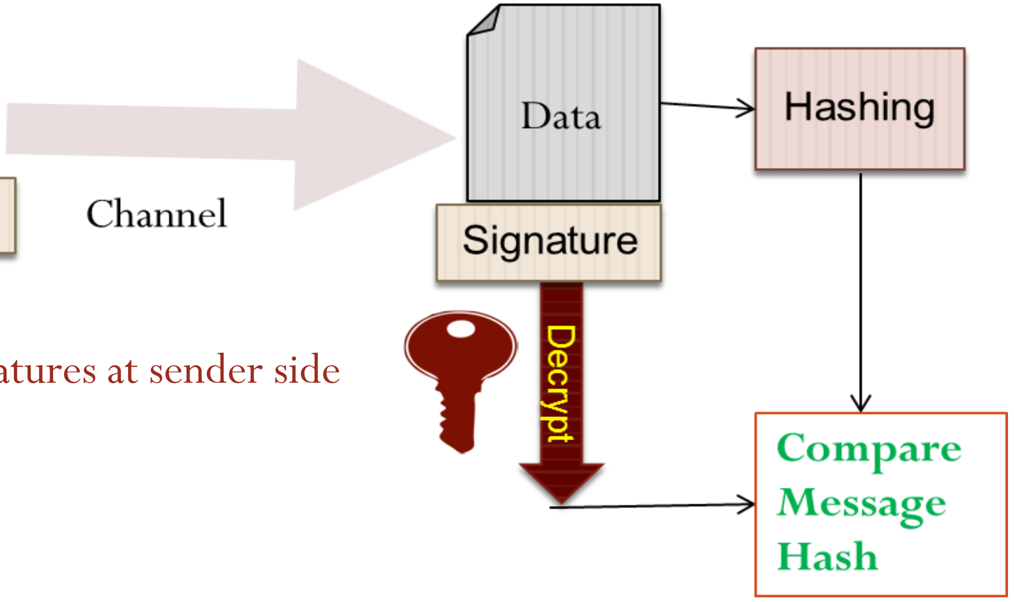
Integrity: Digital Signature Algorithms



Generating hash value



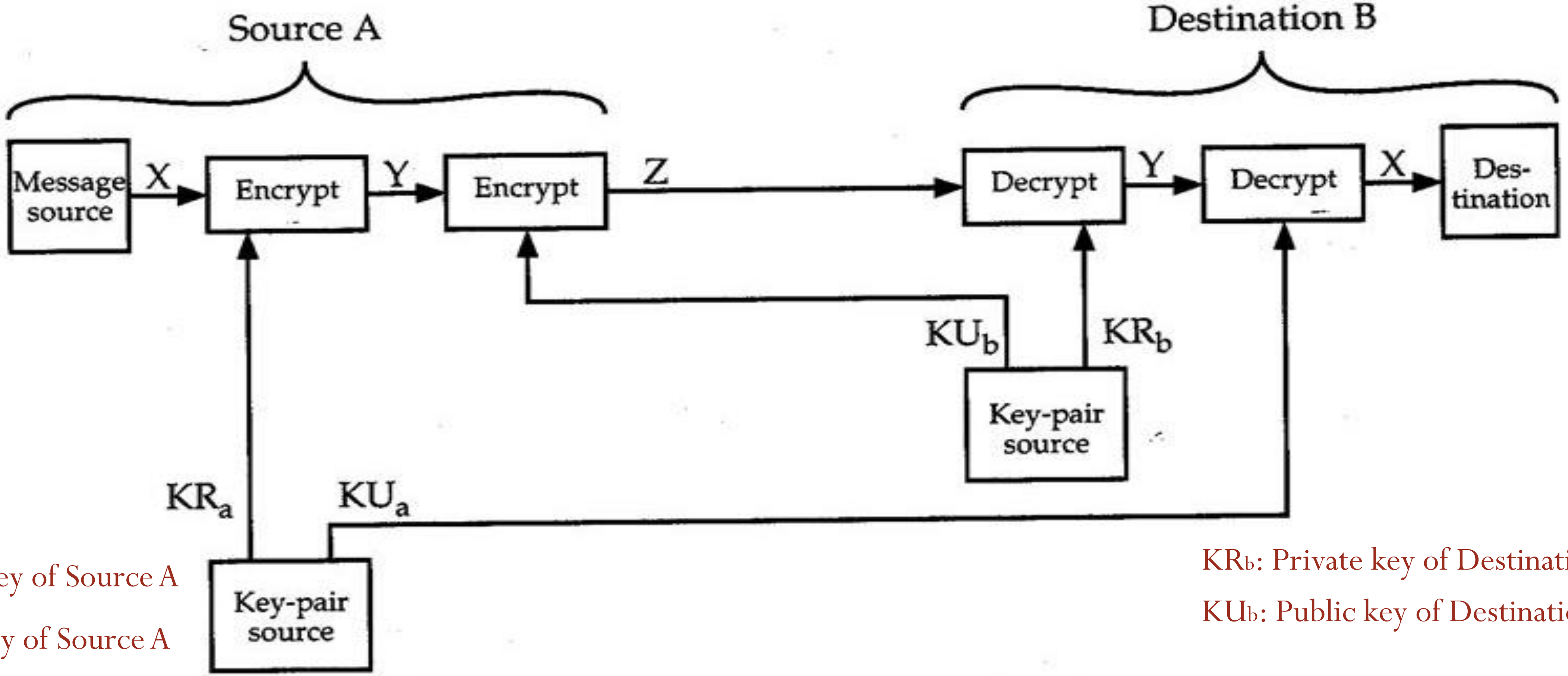
Digital signatures at sender side



Digital signatures at receiver side

Digital signature

Authentication and non-repudiation :Public key cryptography



KRa: Private key of Source A

KUa: Public key of Source A

KRb: Private key of Destination B

KUb: Public key of Destination B

sender side

receiver side

Security Attacks

- Potential attacks means that there is not just one method of meeting a particular security requirement: each of the types of attacks that present a specific threat needs to be countered.

Confidentiality	Integrity	Availability	Non-repudiation
Eves dropping	**Bypassing controls**	**Denial of Service**	**Repudiation**
Traffic analysis	Authorization violation		
	Physical intrusion		
	Integrity violation		
	Intercept/ alter		
	Replay		

Authentication
Man In The Middle
Masquerade